

Intro to AI for Medicine

Lec 7: Case Studies

Last Time

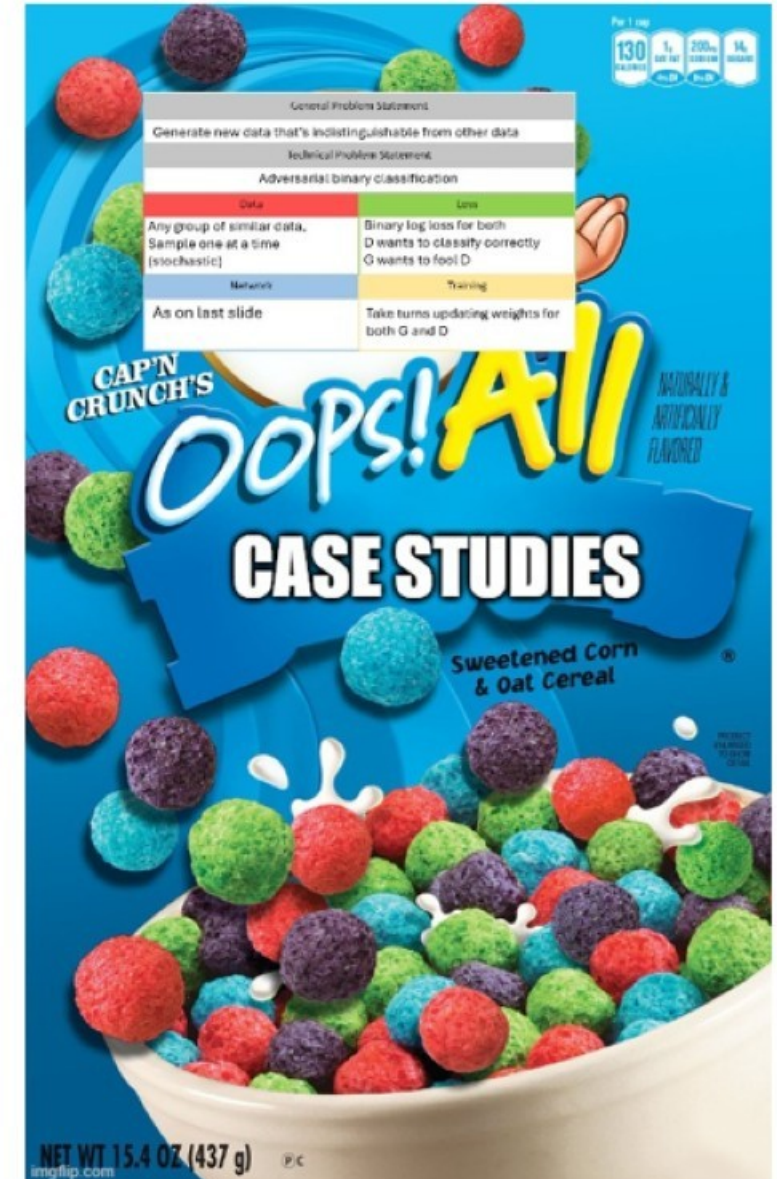
- Discussed various AI networks and their use cases
 - RNN/LSTM
 - Autoencoder
 - GAN
- Mentioned Random Forest methods

By This Point...

- You know the basics of AI
 - Data
 - Loss
 - Network
 - Training
- You know how to break down and interpret how AI systems work, even ones you don't know about

Today

- 4 case studies
- How do you take a paper and understand what's going on



General Problem Statement

In plain English, what are we trying to achieve?

Technical Problem Statement

What is the technical approach to achieving above? Regression, classification, generative, LLM?

Data

What is the sample population?

Type of data - image, chart, time series, multimodal

Augmentations?

Loss

Classically, should match problem type:

- Regression: MSE, Classification: Log loss/Multiclass log loss

What is loss function rewarding? Consider regularization.

Network

Classic neural net or with variations?

autoencoder, GAN, RNN, LSTM, CNN, combo of these. If not sure, what's it closest to?

If brand new, what kind of data can it take in and process well?

Training

What hyperparameters were considered? Should have to do with the network structure

Use of pretrained model weights?

Train/test split—random, cross validation

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Data	Loss
What is the sample population? Type of data - image, chart, time series, multimodal Augmentations?	Classically, should match problem type: - Regression: MSE, Classification: Log loss/Multiclass log loss What is loss function rewarding? Consider regularization.
Network	Training
Classic neural net or with variations? autoencoder, GAN, RNN, LSTM, CNN, combo of these. If not sure, what's it closest to? If brand new, what kind of data can it take in and process well?	What hyperparameters were considered? Should have to do with the network structure Use of pretrained model weights? Train/test split—random, cross validation
Results	
Sensitivity/specificity, PPV/NPV, AU-ROC? Imaging: DICE, Hausdorff Distance Other measures of success? Clinical relevance? Risk of bias?	

Case Study 1



Letter | Published: 07 January 2019

Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network

[Awni Y. Hannun](#) , [Pranav Rajpurkar](#), [Masoumeh Haghpanahi](#), [Geoffrey H. Tison](#), [Codie Bourn](#), [Mintu P. Turakhia](#) & [Andrew Y. Ng](#)

[Nature Medicine](#) **25**, 65–69 (2019) | [Cite this article](#)

73k Accesses | **2348** Citations | **477** Altmetric | [Metrics](#)

<https://www.nature.com/articles/s41591-018-0268-3>



General Problem Statement

Classify ZioPatch readings into one of many types of arrhythmia

Technical Problem Statement

Multiclass classification of EKG images into classes using convolutional/DNN

Data

91,232 single-lead ECGs from 53,549 patients
Annotation of ground truth by cardiologists
The atrial fibrillation class combined atrial fibrillation and atrial flutter. The AVB class combined both type 2 second-degree AVB (Mobitz II/Hay) and third-degree AVB.

Loss

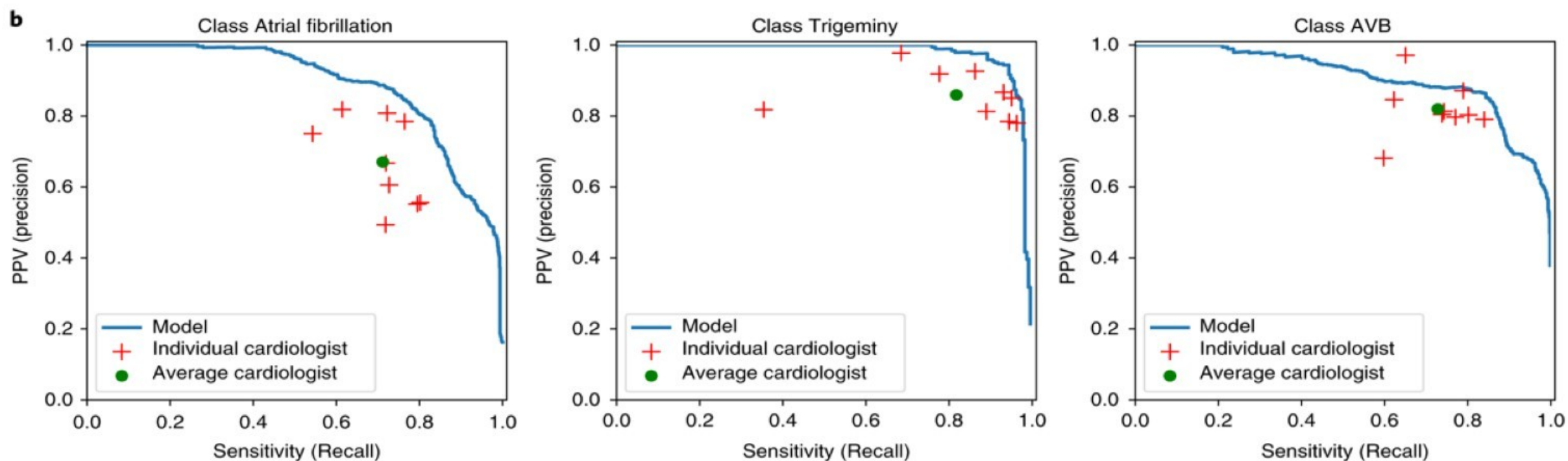
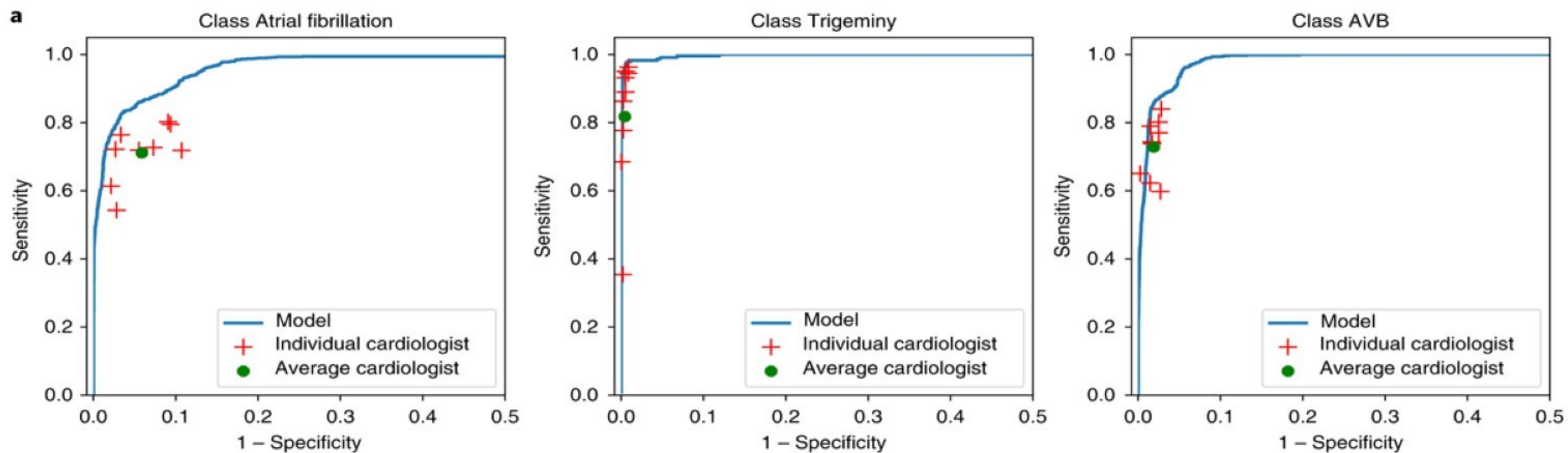
Rhythm predictions every 1.28 seconds
Compared ground truth and algorithm at every 1.28 second interval, as well as diagnoses across whole recording time

Network

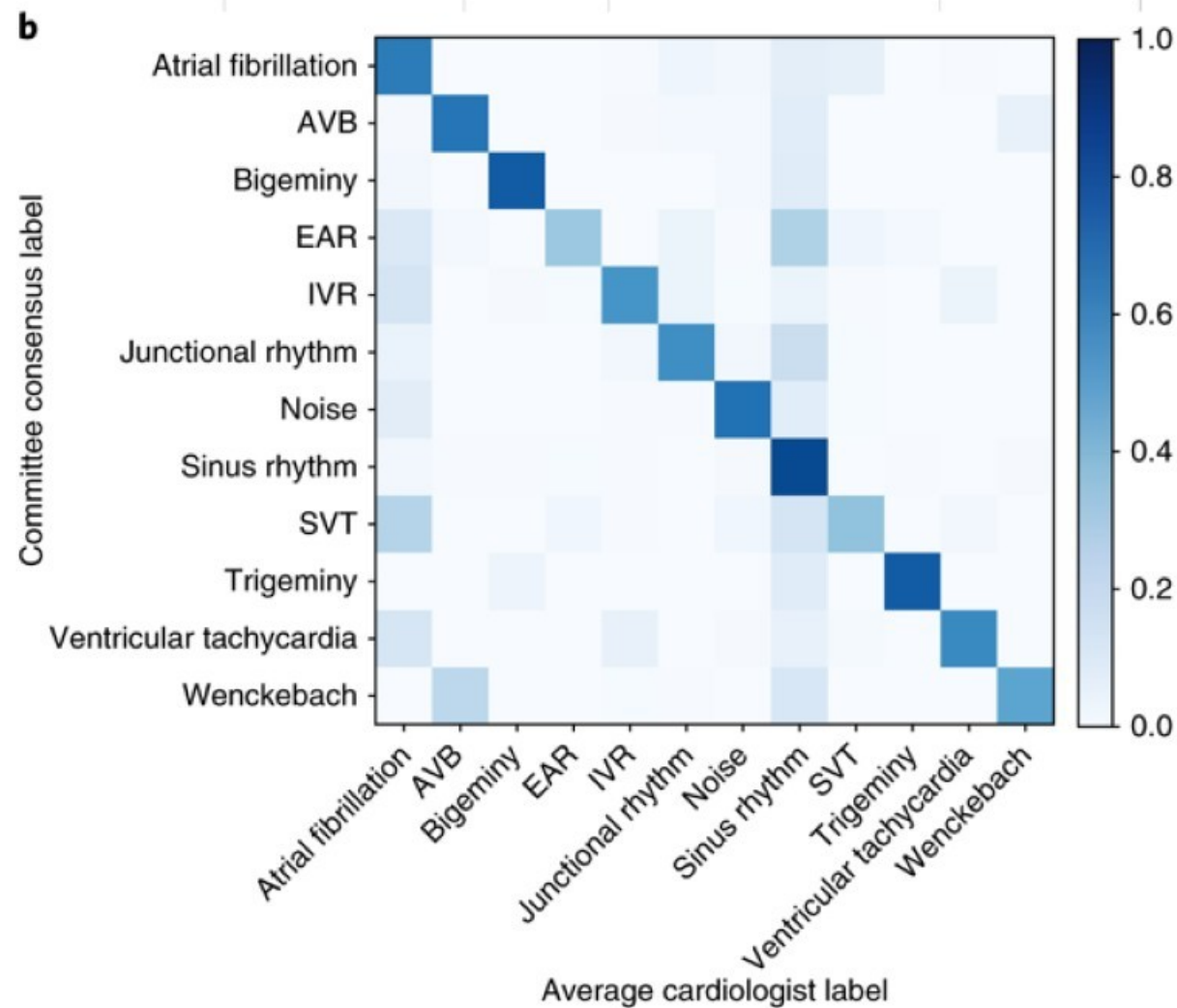
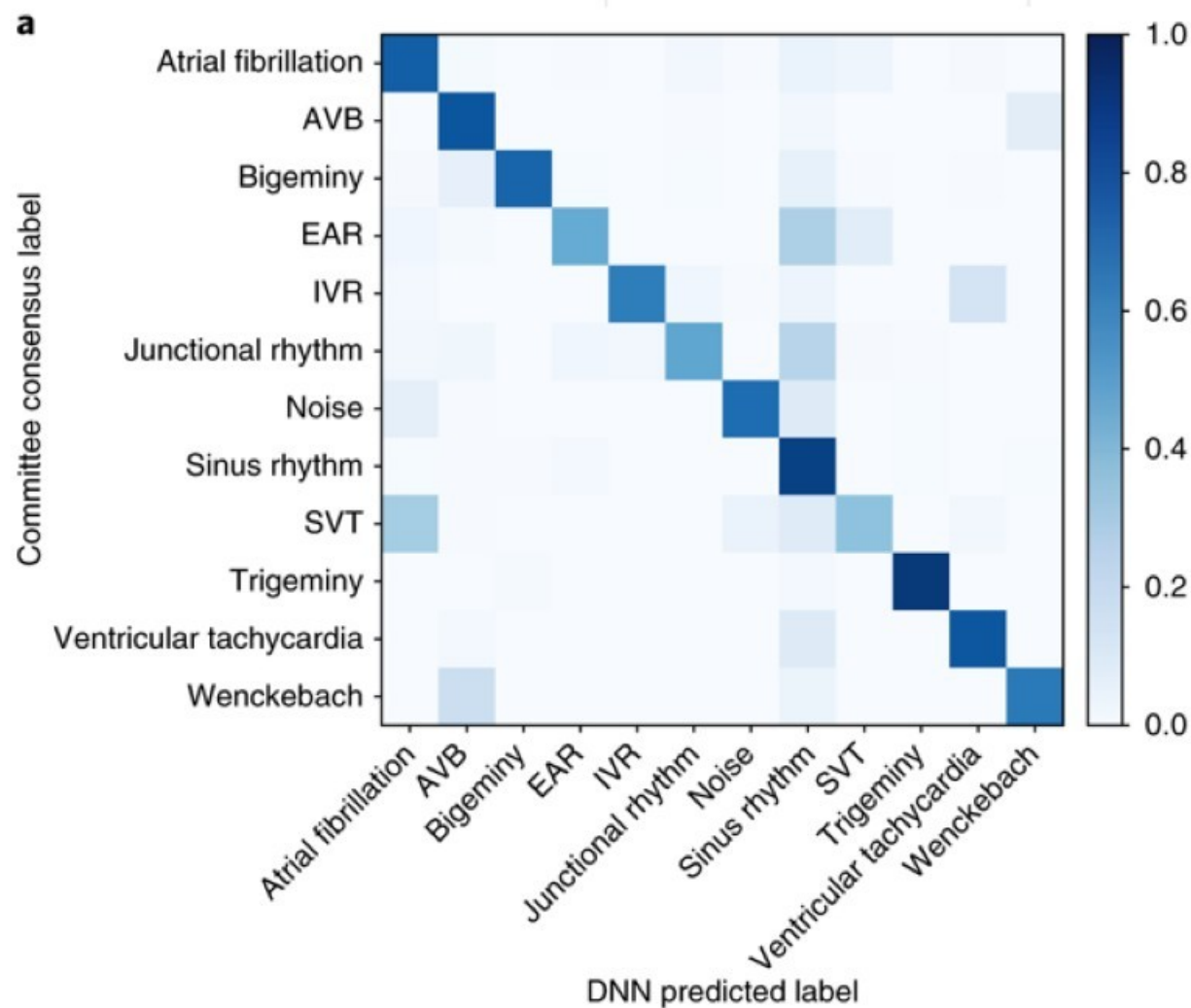
Convolutional DNN, which decreases dimensionality layer after layer
Batch norm, ReLU activation, Dropout

Training

Held out records from a random 10% of the training dataset patients for use as a development dataset to perform DNN hyper-parameter tuning.
Cardiologist data was test, else train was by annotator



	Algorithm AUC (95% CI) ^a		Algorithm F_1 ^b		Average cardiologist F_1	
	Sequence ^a	Set ^b	Sequence	Set	Sequence	Set
Frequency-weighted average	0.978	0.977	0.807	0.837	0.753	0.780



Key: **tp** = True Positive, **tn** = True Negative, **fp** = False Positive, **fn** = False Negative

Metric Name	Metric Forumla	Code	When to use
Accuracy	Accuracy = $\frac{tp + tn}{tp + tn + fp + fn}$	<code>tf.keras.metrics.Accuracy()</code> or <code>sklearn.metrics.accuracy_score()</code>	Default metric for classification problems. Not the best for imbalanced classes.
Precision PPV	Precision = $\frac{tp}{tp + fp}$	<code>tf.keras.metrics.Precision()</code> or <code>sklearn.metrics.precision_score()</code>	Higher precision leads to less false positives.
Recall Sensitivity	Recall = $\frac{tp}{tp + fn}$	<code>tf.keras.metrics.Recall()</code> or <code>sklearn.metrics.recall_score()</code>	Higher recall leads to less false negatives.
F1-score	F1-score = $2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$	<code>sklearn.metrics.f1_score()</code>	Combination of precision and recall, usually a good overall metric for a classification model.
Confusion matrix	NA	Custom function or <code>sklearn.metrics.confusion_matrix()</code>	When comparing predictions to truth labels to see where model gets confused. Can be hard to use with large numbers of classes.

Case Study 2

Article | [Open access](#) | Published: 28 December 2024

Explainable artificial intelligence for stroke prediction through comparison of deep learning and machine learning models

[Khadijeh Moulaei](#), [Lida Afshari](#), [Reza Moulaei](#), [Babak Sabet](#), [Seyed Mohammad Mousavi](#) & [Mohammad Reza Afrash](#) 

[Scientific Reports](#) **14**, Article number: 31392 (2024) | [Cite this article](#)

9169 Accesses | **6** Citations | **1** Altmetric | [Metrics](#)

<https://www.nature.com/articles/s41598-024-82931-5#Sec11>

General Problem Statement

Predict if a patient will have a stroke

Technical Problem Statement

Classification task using multiple models

Data

Dataset comprising 663 records from one Iranian hospital
401 cases, 262 controls
Predicting stroke in what timeframe? Unclear
53 important features influencing stroke in patients were identified by two neurologists.
Variables with missingness less than 30% were imputed,

Loss

Varies based on network

Network

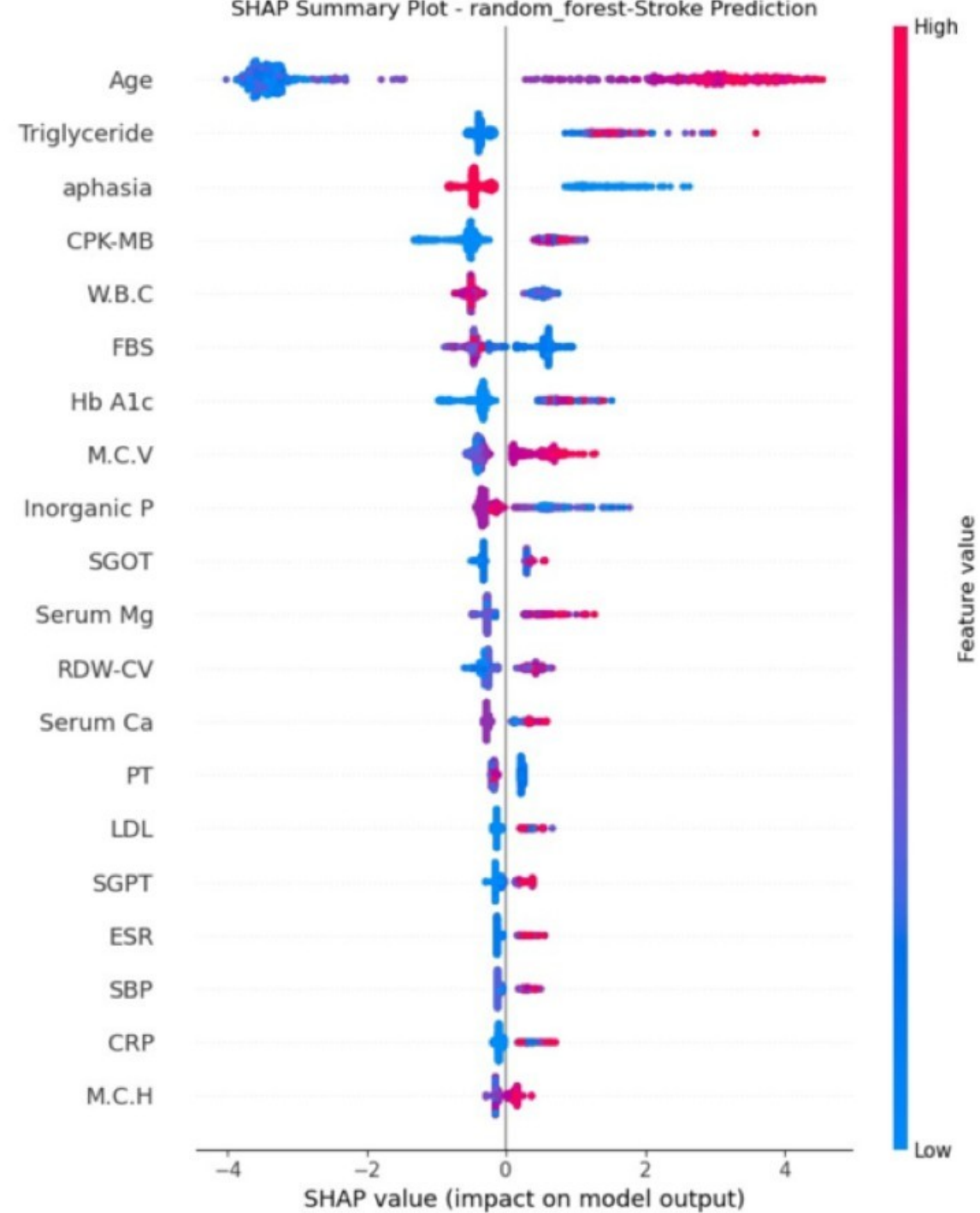
A total of eight established ML (SVM, XGB, KNN, RF) and DL (DNN, FNN, **LSTM?**, **CNN?**) models were utilized to predict stroke.

Training

10-fold cross-validation approach

Models	Sensitivity (%)	Specificity (%)	Accuracy (%)	F1-score (%)	ROC (%)
Performance of each deep learning models					
CNN	96.0	87.0	92.0	91.0	96.0
LSTM	96.15	80.0	89.47	87.34	97.0
DNN	93.0	93.0	95.0	93.0	95.0
FNN	94.0	96.0	96.0	95.0	98.0
Performance of each machine learning models					
RF	99.9	100	99.0	99.0	99.0
KNN	70.0	94.0	86.0	80.0	83.0
XGB	94.0	95.0	95.0	95.0	95.0
SVM	29.0	94.0	71.0	44.0	64.0

Num	ML & DP models	Hyper-Parameters
1	CNN	Conv1D (filters = 64, kernel_size = 5, MaxPoolingD (pool_size = 3), Dense (2)
2	LSTM	LSTM Units = 64, Activation Function= 'sigmoid', Optimizer='adam', Loss Function='binary_crossentropy', Epochs = 20, Batch Size = 128.
3	DNN	Layers and Neurons = 84, 42, 1, Activation Functions='relu', 'relu', 'sigmoid',Optimizer='adam', Loss Function= 'binary_crossentropy', Epochs = 20, Batch Size = 128.
4	FNN	Hidden Layer Size= (128,), Max Iterations = 2000, Random State = 42, Activation Function='relu', Solver='adam', Learning Rate='constant', Batch Size='128'.
5	RF	n_estimators = 400, criterion=' entropy', max_depth = None, min_samples_split = 2, min_samples_leaf = 1, max_features='auto', bootstrap = True, oob_score = False, random_state = 42.
6	KNN	Test_size = 0.2, Random_state: 42, CV = 10
7	XGB	learning_rate = 0.05, n_estimators = 500, max_depth = 10, min_child_weight = 5, subsample = 0.9, colsample_bytree = 0.8, gamma = 0, reg_lambda = 0.
8	SVM	Probability='True', C=' 5.0', kernel = 'rbf', degree=' 0 ' gamma = 0.1



Takeaway

- Promising paper, but needs to be repeated
- “Too good to be true” esp. in the context of authors making questionable network decisions

From another paper

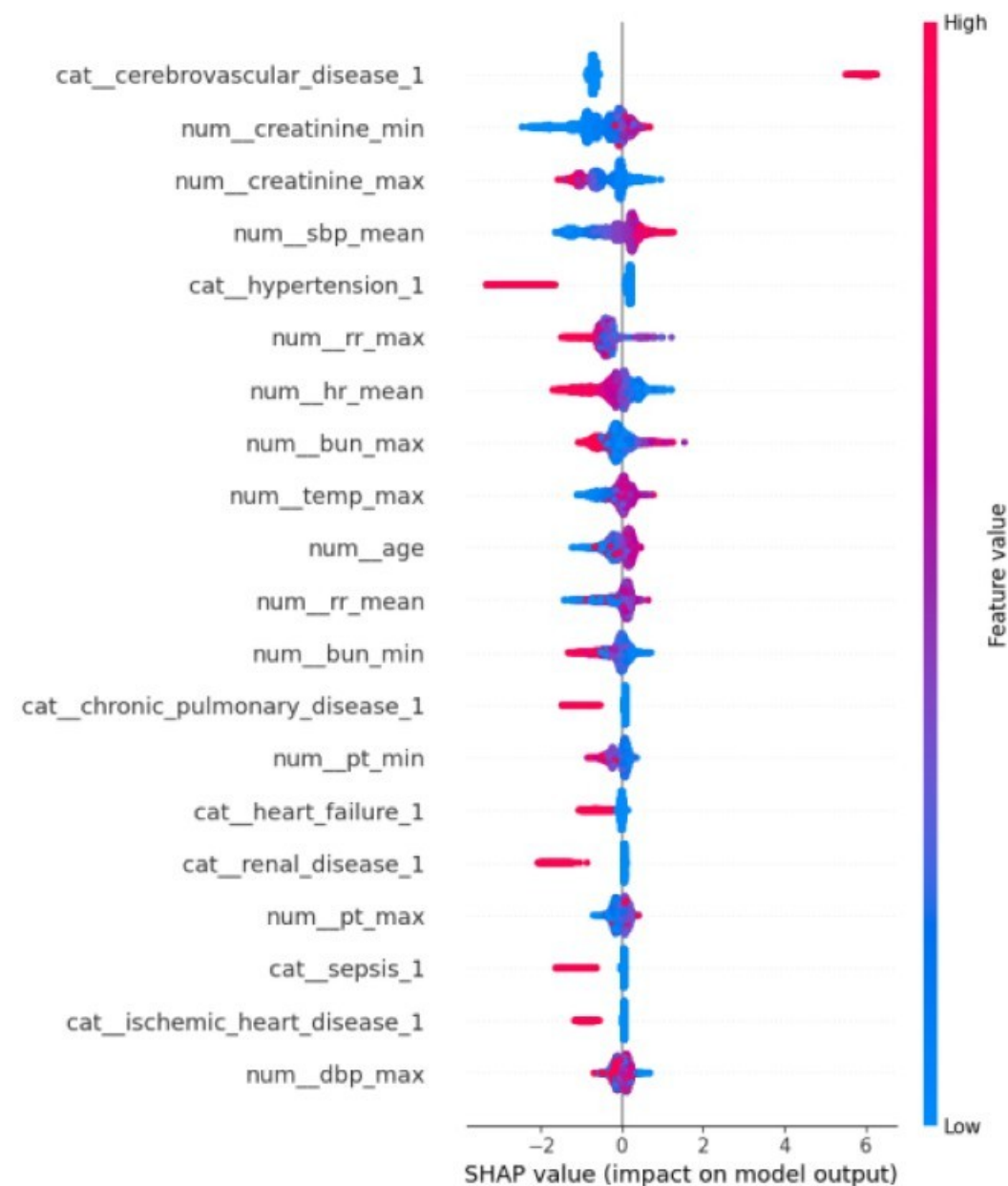
Table 5. Training Set Performance Comparison of Classifiers

Model	AUROC	Accuracy	F1 Score	Sensitivity	Specificity	PPV	NPV
Logistic Regression	0.8612	0.7272	0.4450	0.8125	0.7140	0.3064	0.9608
Naive Bayes	0.6767	0.4100	0.3072	0.9724	0.3225	0.1824	0.9869
SVM	0.8945	0.7657	0.4975	0.8620	0.7507	0.3497	0.9722
KNN	0.9989	0.8138	0.5910	0.9999	0.7848	0.4194	1.0000
Gradient Boosting	0.8761	0.8579	0.5524	0.6515	0.8900	0.4794	0.9426
XGBoost	0.9925	0.9706	0.8818	0.8155	0.9947	0.9598	0.9720
LightGBM	0.9580	0.9359	0.7194	0.6108	0.9864	0.8749	0.9422
CatBoost	0.9860	0.9599	0.8294	0.7256	0.9963	0.9681	0.9589

Table 6. Test Set Performance Comparison of Classifiers

Model	AUROC	Accuracy	F1 Score	Sensitivity	Specificity	PPV	NPV
Logistic Regression	0.8562	0.7249	0.4412	0.8071	0.7121	0.3036	0.9596
Naive Bayes	0.6739	0.4089	0.3057	0.9676	0.3221	0.1815	0.9846
SVM	0.8559	0.7480	0.4590	0.7946	0.7407	0.3228	0.9587
KNN	0.7830	0.7065	0.4030	0.7361	0.7019	0.2775	0.9448
Gradient Boosting	0.8565	0.8493	0.5267	0.6231	0.8845	0.4563	0.9378
XGBoost	0.8822	0.9093	0.5963	0.4979	0.9733	0.7433	0.9257
LightGBM	0.8774	0.9029	0.5568	0.4534	0.9728	0.7216	0.9196
CatBoost	0.8868	0.9112	0.5817	0.4585	0.9817	0.7957	0.9209

<https://arxiv.org/pdf/2506.03209>



Case Study 3



BoneMRI[®]

**IN THE ASSESSMENT OF SPINAL
FRACTURES**

Soft tissue information and bony details from
a single radiation-free scan.

Claims



100% fracture detection

in the cervical spine when combined with standard MRI sequences, and 93% detection using BoneMRI alone³



Findings beyond what CT shows

BoneMRI combined with T2W imaging, reveals soft tissue injuries missed by CT, like ligament damage and spinal cord injuries, potentially impacting clinical decision making.³



All-in-one exam

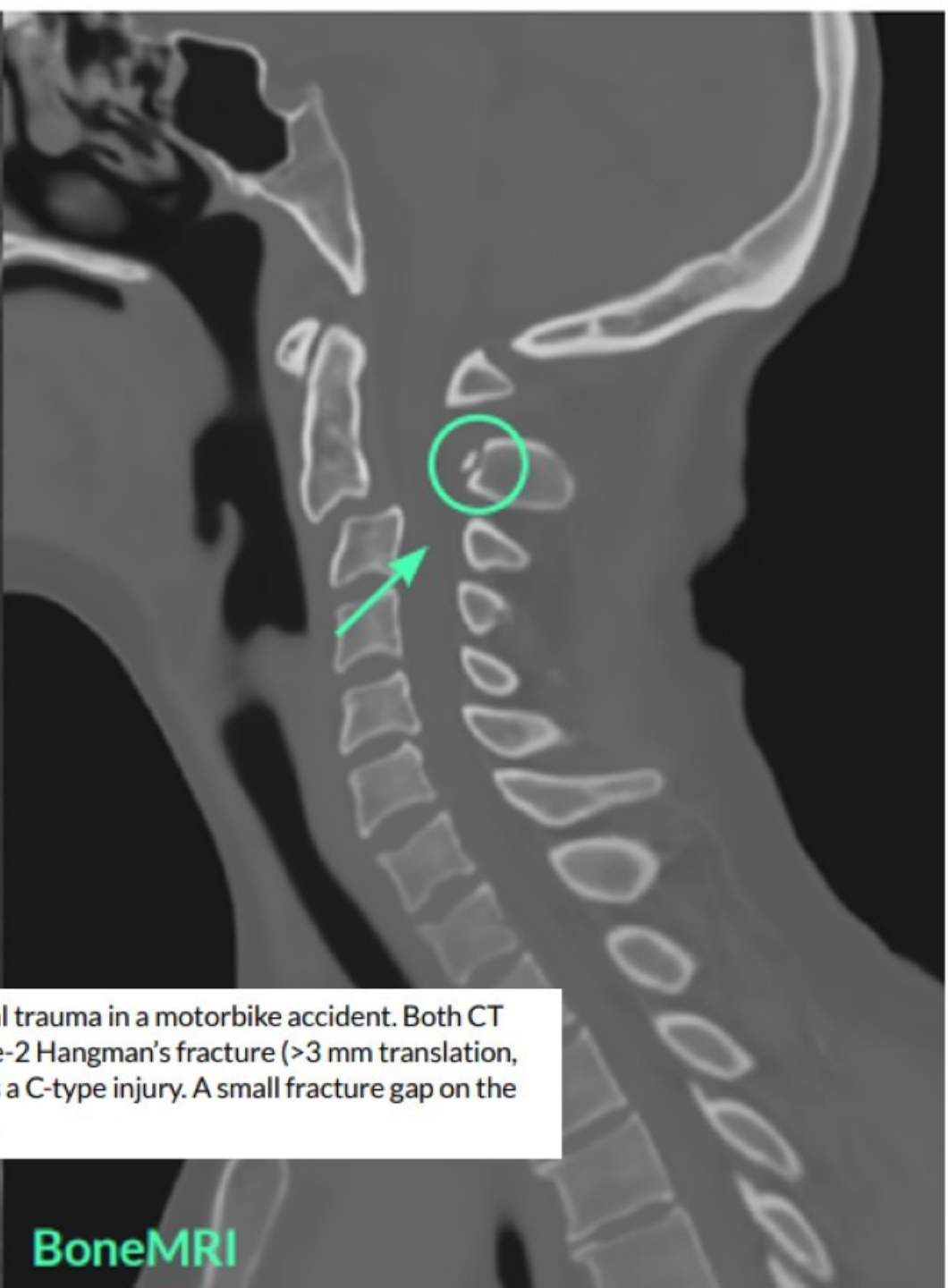
Bone imaging with soft tissue evaluation in a single, radiation-free MRI scan, eliminating the need for CT.



Streamlined workflow

Delivers all key data in one session, reducing imaging steps and boosting diagnostic confidence.

Business Pros/Cons



A 45-year-old female sustained blunt cervical trauma in a motorbike accident. Both CT and BoneMRI showed a Levine Edwards type-2 Hangman's fracture (>3 mm translation, no angulation). According to AO Spine, this is a C-type injury. A small fracture gap on the C2 spinous process was visible in both scans.

CT

BoneMRI

General Problem Statement

Generate an anatomically and visually accurate synthetic CT from an MRI

Technical Problem Statement

Voxel-wise regression of MRI intensities to CT intensities (Hounsfield units)

Data

MRI (many phases overlaid) + CT
18-80 traumatic C-spine @ 2 hospitals w/o
contraindication
Ground truth for fracture determined by 3x spine
surgeons, 2x radiologists

Loss

The similarity between the CT and sCT was assessed
using 5 metrics: mean error (ME), mean absolute error
(MAE), Dice similarity coefficient (DSC), surface
distance, and peak SNR
L1 regularization (lasso) (emphasizing sparse network)

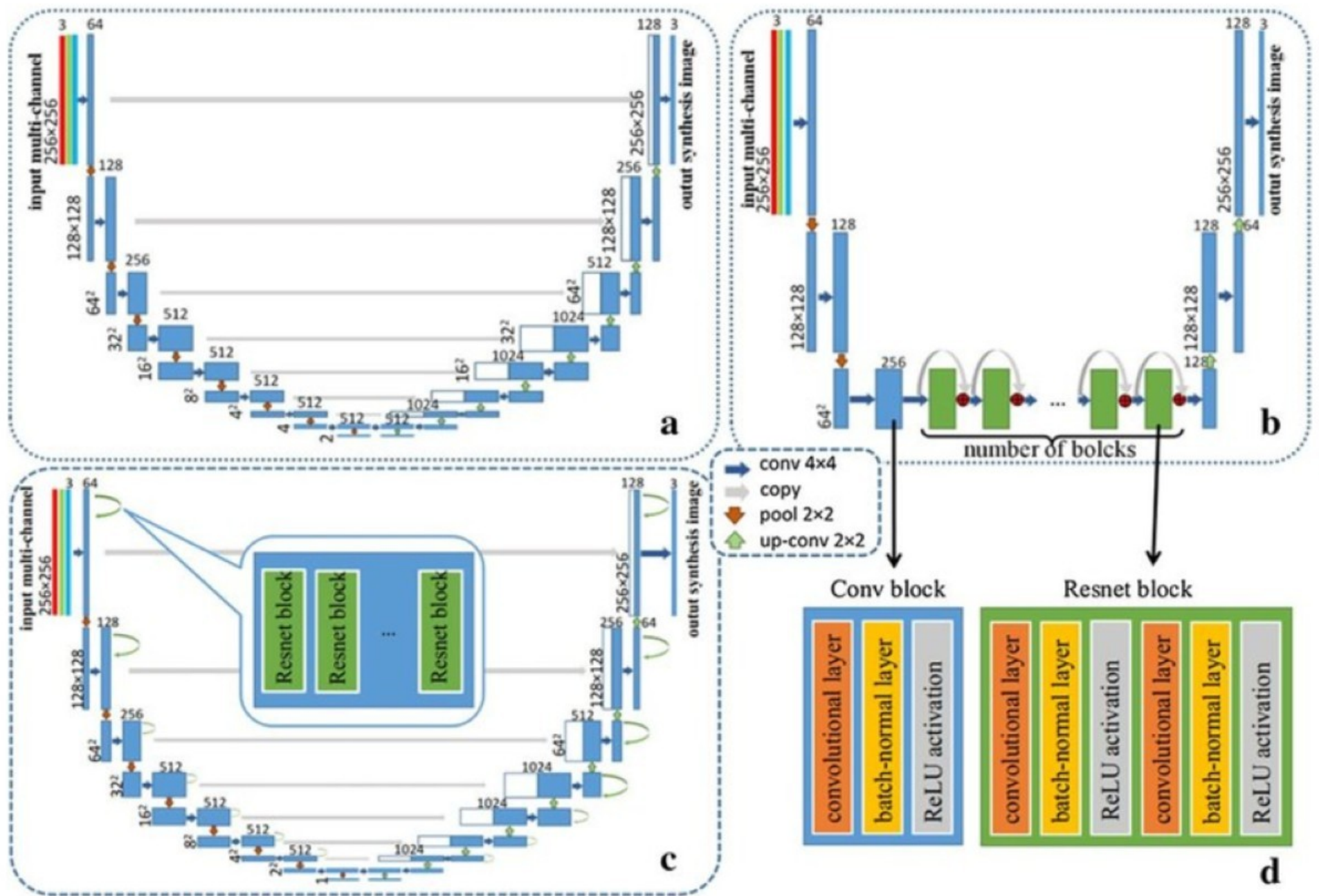
Network

Next slide

Training

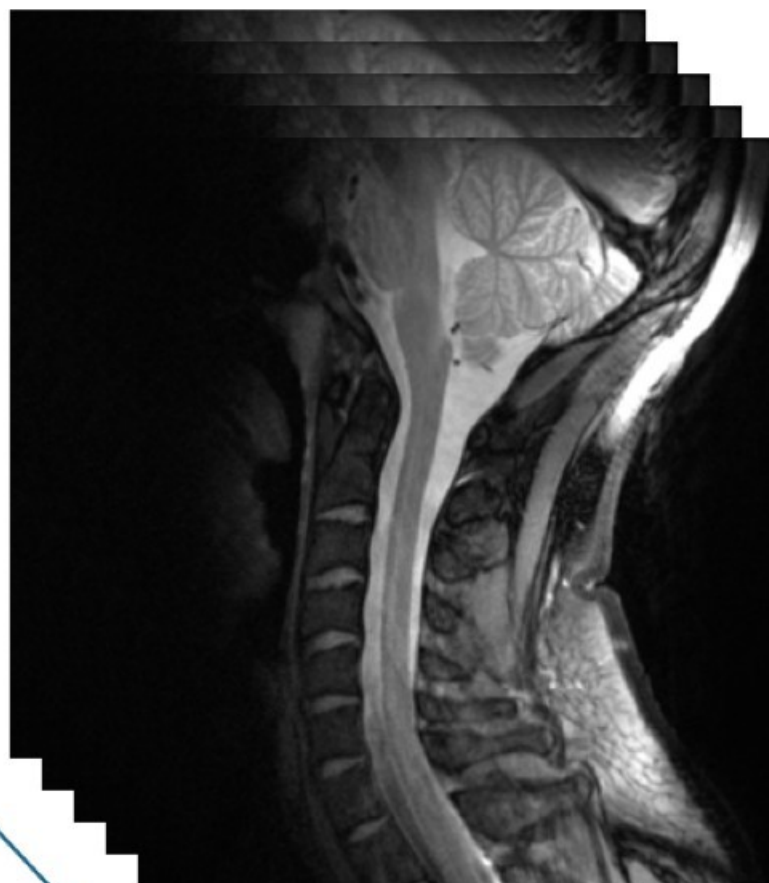
3-fold cross-validation approach
Pre-training registration (alignment)

Visual autoencoder
known as a U-net



Generator network. a U-net. b ResNet. c ResU-net. d Details of blocks

Input



Sequences

Height

Width

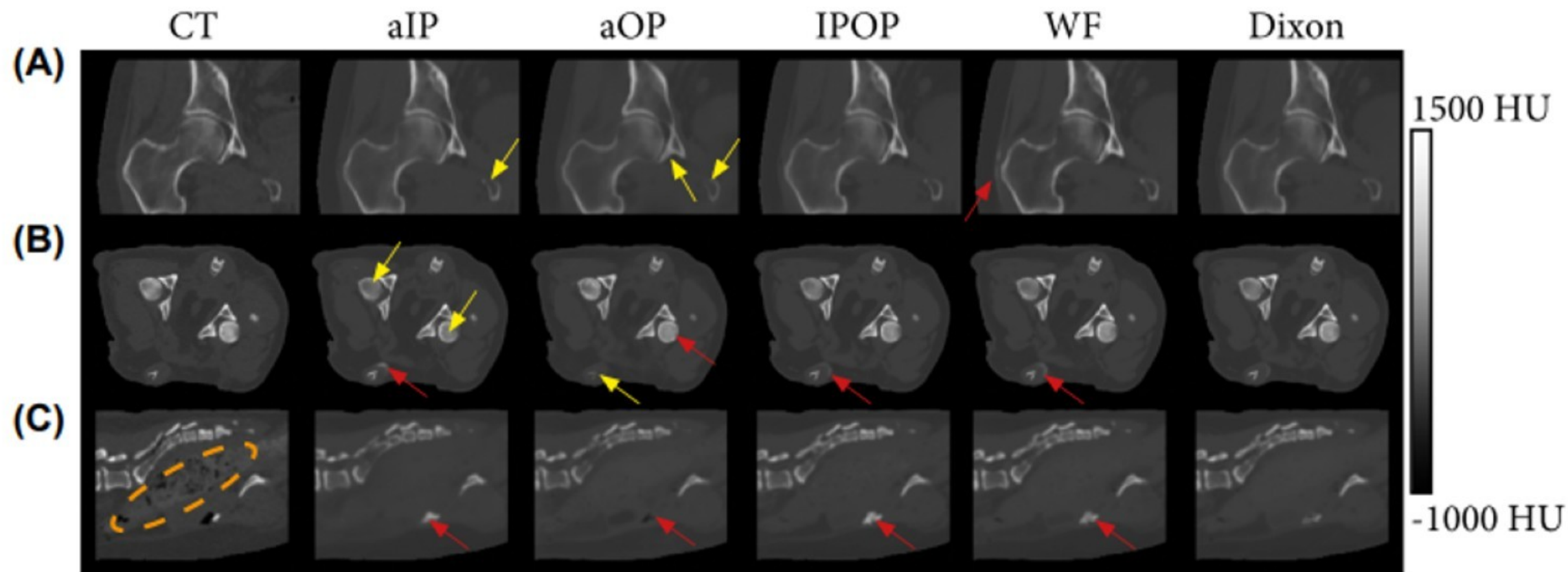
Output



Spin: -90
Tilt: -2

L

Multi Sequence Input Is Critical



Results

	Sensitivity sCT	
	Type A, B, C and C1, C2 fractures	All fracture types (including F-Type)
sCT	97.3%	92.7%
sCT and STIR/TIRM/T2	100%	100%

Parameter	Intermodal agreement (sCT vs. CT)			Inter-reader agreement (ICC)	
	Mean absolute difference	ICC	<i>p</i> -value	sCT	CT
AVH	0.3 mm	0.95	< 0.001	0.87; <i>p</i> < 0.001	0.88; <i>p</i> < 0.001
PVH	0.3 mm	0.98	< 0.001	0.88; <i>p</i> < 0.001	0.87; <i>p</i> < 0.001
VBA	1.15°	0.97	< 0.001	0.78; <i>p</i> < 0.001	0.76; <i>p</i> < 0.001
SK	0.51°	0.83	< 0.001	0.77; <i>p</i> < 0.001	0.93; <i>p</i> < 0.001

Should you use this?

Case Study 4

LEARNING TO DIAGNOSE WITH LSTM RECURRENT NEURAL NETWORKS

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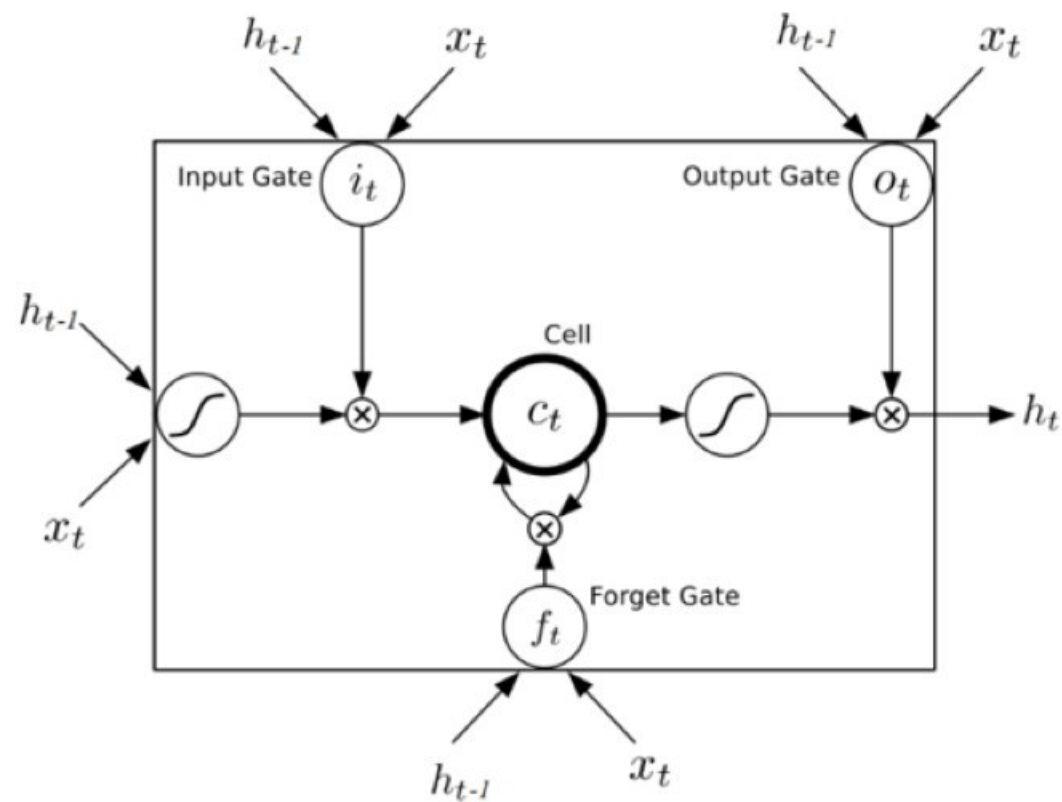
David C. Kale ^{*‡}

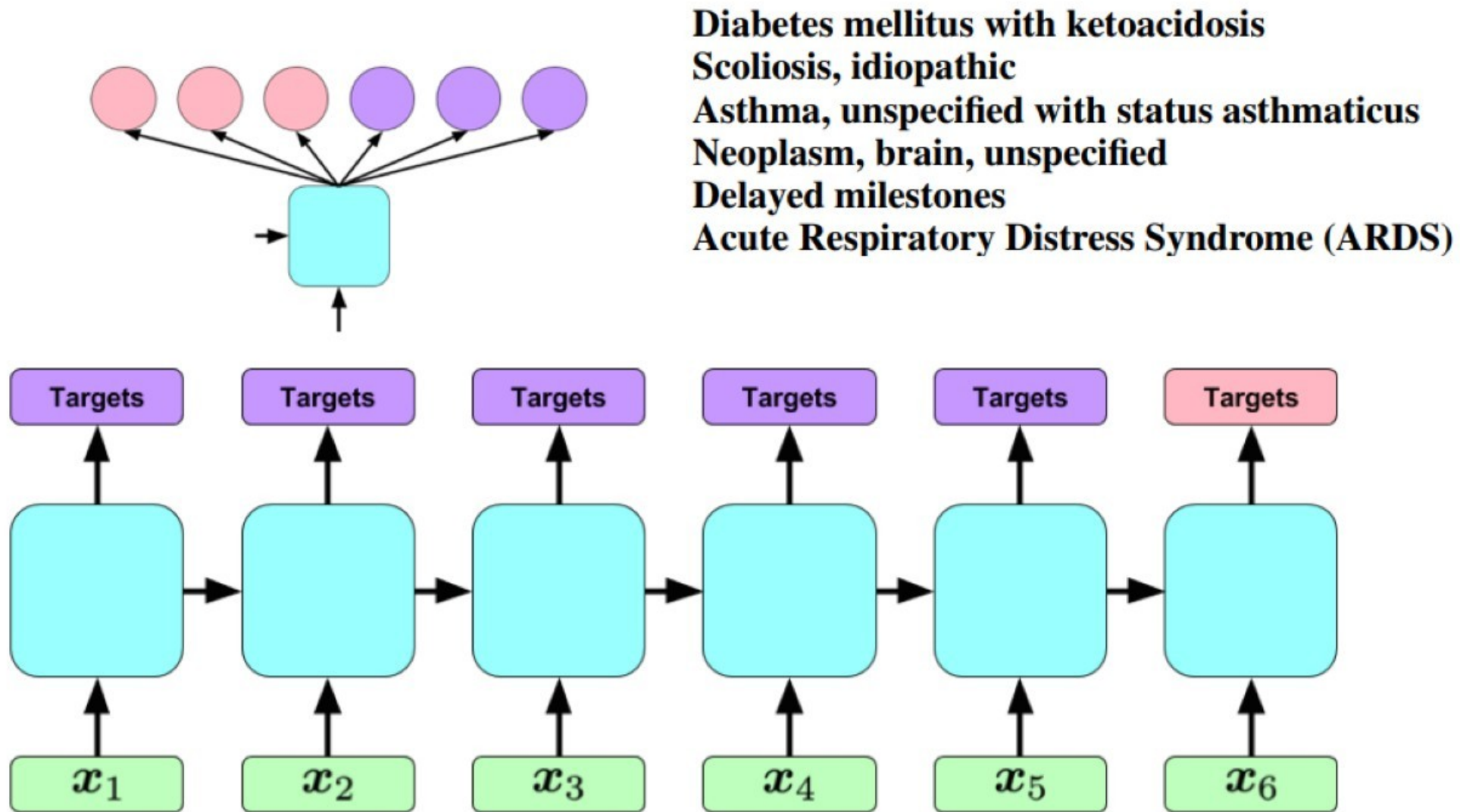
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Long Short Term Memory





General Problem Statement

Predict PICU diagnoses days/weeks before they happen

Technical Problem Statement

Multi- single class classification of time series data using LSTM model, compare to existing models

Data

10, 401 PICU episodes, each a multivariate time series of 13 variables

Single academic hospital PICU (LA Childrens)

Scaling, imputing

Output: diagnostic codes from hospital stay, as determined by physicians

Loss

We use log loss as the loss function at each output.

Network

Last slide

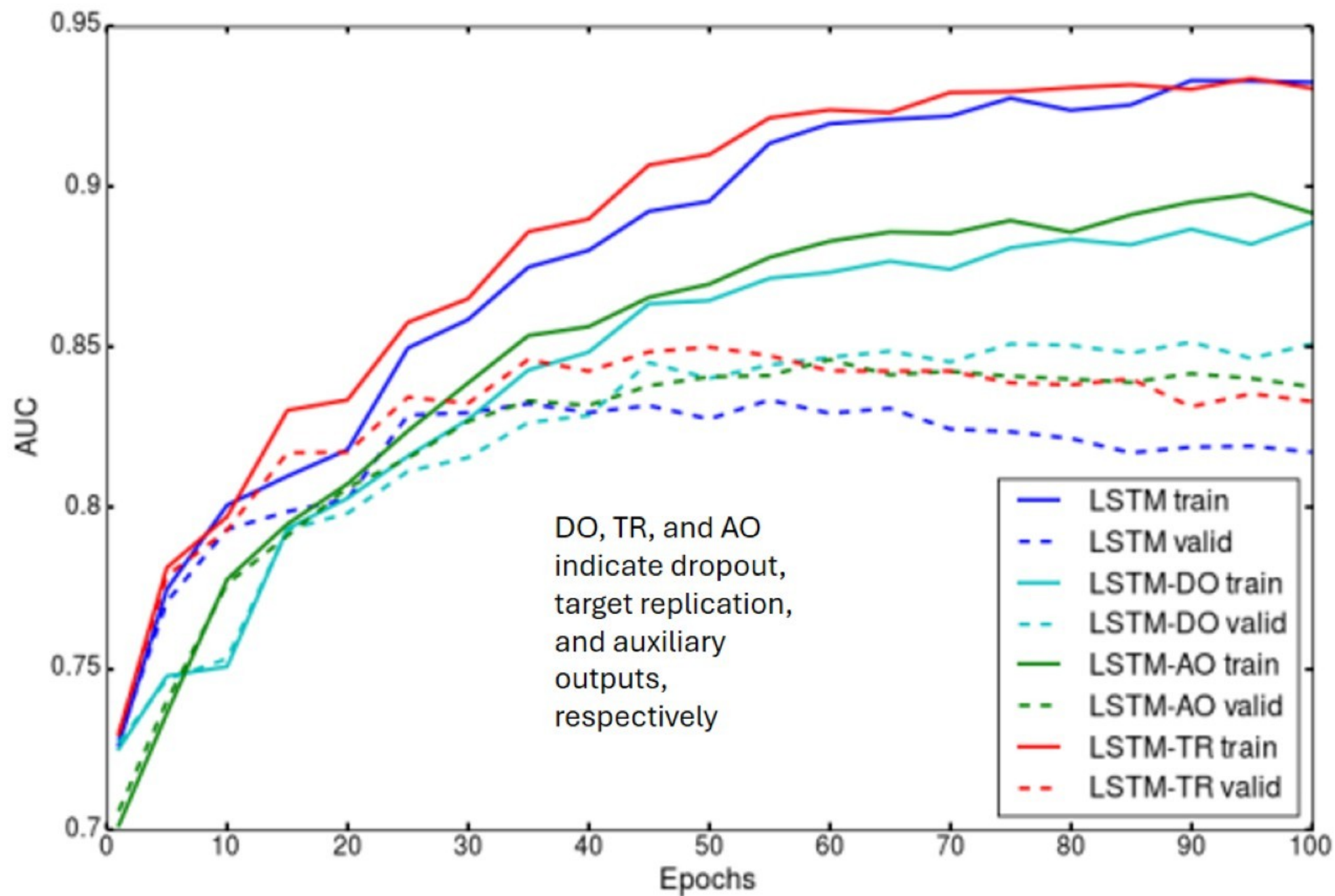
Recall that network allows us to maintain long-term time dependencies

Regularization + Dropout

Training

100 epochs stochastic GD

L2 regularization (prevent overweighing)



Classification performance for 128 ICU phenotypes

Model	Micro AUC	Macro AUC	Micro F1	Macro F1	Prec. at 10
Base Rate	0.7128	0.5	0.1346	0.0343	0.0788
Log. Reg., First 6 + Last 6	0.8122	0.7404	0.2324	0.1081	0.1016
Log. Reg., Expert features	0.8285	0.7644	0.2502	0.1373	0.1087
MLP, First 6 + Last 6	0.8375	0.7770	0.2698	0.1286	0.1096
MLP, Expert features	0.8551	0.8030	0.2930	0.1475	0.1170
LSTM Models with two 64-cell hidden layers					
LSTM	0.8241	0.7573	0.2450	0.1170	0.1047
LSTM, AuxOut (Diagnoses)	0.8351	0.7746	0.2627	0.1309	0.1110
LSTM-AO (Categories)	0.8382	0.7748	0.2651	0.1351	0.1099
LSTM-TR	0.8429	0.7870	0.2702	0.1348	0.1115
LSTM-TR-AO (Diagnoses)	0.8391	0.7866	0.2599	0.1317	0.1085
LSTM-TR-AO (Categories)	0.8439	0.7860	0.2774	0.1330	0.1138
LSTM Models with Dropout (probability 0.5) and two 128-cell hidden layers					
LSTM-DO	0.8377	0.7741	0.2748	0.1371	0.1110
LSTM-DO-AO (Diagnoses)	0.8365	0.7785	0.2581	0.1366	0.1104
LSTM-DO-AO (Categories)	0.8399	0.7783	0.2804	0.1361	0.1123
LSTM-DO-TR	0.8560	0.8075	0.2938	0.1485	0.1172
LSTM-DO-TR-AO (Diagnoses)	0.8470	0.7929	0.2735	0.1488	0.1149
LSTM-DO-TR-AO (Categories)	0.8543	0.8015	0.2887	0.1446	0.1161
LSTM-DO-TR (Linear Gain)	0.8480	0.7986	0.2896	0.1530	0.1160
Ensembles of Best MLP and Best LSTM					
Mean of LSTM-DO-TR & MLP	0.8611	0.8143	0.2981	0.1553	0.1201
Max of LSTM-DO-TR & MLP	0.8643	0.8194	0.3035	0.1571	0.1218

DO, TR, and AO indicate dropout, target replication, and auxiliary outputs, respectively